

Exam Prep AI - Mock Question Paper

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Page 1 of 8

Course Code: UCEST105

Course Name: ALGORITHMIC THINKING WITH PYTHON

Max. Marks: 60

Duration: 2 hours 30 minutes

PART A

(Answer all questions. Each question carries 3 marks)

CO Marks

1

A student is trying to solve a word search puzzle. The puzzle consists of a 10x10 grid of letters, and the student is trying to find the word "COMPUTER". The student has a hint that the word is located in the top-left to bottom-right diagonal of the grid. Describe the algorithm the student should use to find the word.

CO-1

(3)

2

Write a Python function to check if a given string is a palindrome.

CO-1

(3)

3

Write a Python program to find the maximum value in a list of numbers.

CO-3

(3)

4

A bank is offering a promotion where customers can earn a 10% bonus on their deposits. If a customer deposits \$100, they will earn a bonus of \$10, making their total balance \$110. Write an algorithm to calculate the total balance after a customer deposits a certain amount of money.

CO-4

(3)

5

Describe the difference between a recursive and an iterative algorithm.

CO-1

(3)

6

Draw a flowchart to find the maximum value in a list of numbers.

CO-2

(3)

7

Write a Python program to find the average value in a list of numbers.

CO-3

(3)

8

A company is offering a discount of 15% on all items priced above \$50. Write an algorithm to calculate the discounted price of an item.

CO-4

(3)

9

Describe the concept of Big O notation and its importance in algorithm analysis.

CO-1

(3)

10

Draw a flowchart to find the average value in a list of numbers.

CO-2

(3)

11

Write a Python program to find the sum of all numbers in a list.

CO-3

(3)

12

A student is trying to solve a math problem that requires finding the sum of all numbers from 1 to 100. Write an algorithm to solve the problem.

CO-4

(3)

13

Describe the difference between a tree and a graph data structure.

CO-1

(3)

14

Draw a flowchart to find the sum of all numbers from 1 to 100.

CO-2

(3)

15

Write a Python program to find the maximum value in a list of strings.

CO-3

(3)

16

A company is offering a promotion where customers can earn a 20% bonus on their purchases. Write an algorithm to calculate the total bonus earned by a customer.

CO-4

(3)

PART B

(Answer any one full question from each module, each question carries 9 marks)

Module -1

17

Explain how the divide and conquer approach can be used to solve the merge sort problem.

CO-1

(9)

18

Explain how the greedy approach can be used to solve the coin change problem.

CO-1

(9)

19

Explain how the dynamic programming approach can be used to solve the knapsack problem.

CO-1

(9)

20

Explain how the backtracking approach can be used to solve the N-Queen problem.

CO-1

(9)

Module -2

21

Write a pseudocode to implement a stack data structure.

CO-2

(9)

22

Write a pseudocode to implement a queue data structure.

CO-2

(9)

23

Write a pseudocode to implement a linked list data structure.

CO-2

(9)

24

Write a pseudocode to implement a binary search tree data structure.

CO-2

(9)

Module -3

25

Write a Python program to implement the bubble sort algorithm.

CO-3

(9)

26

Write a Python program to implement the insertion sort algorithm.

CO-3

(9)

27

Write a Python program to implement the selection sort algorithm.

CO-3

(9)

28

Write a Python program to implement the quicksort algorithm.

CO-3

(9)

Module -4

29

Explain how the divide and conquer approach can be used to solve the closest pair problem.

CO-4

(9)

30

Explain how the greedy approach can be used to solve the activity selection problem.

CO-4

(9)

31

Explain how the dynamic programming approach can be used to solve the longest common subsequence problem.

CO-4

(9)

32

Explain how the backtracking approach can be used to solve the Hamiltonian path problem.

CO-4

(9)